ROUSHAN KUMAR

Python Developer | FastAPI Microservices, Autonomous Automation & Data Pipelines

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PROFESSIONAL SUMMARY

Python Software Developer and Creator of AstreWork, specializing in FastAPI microservice architectures, automated headless browser pipelines, and AST parser sandboxes. Proven ability to build self-healing execution engines with subprocess isolation, intercepting unauthorized system calls in real time. Experienced in distributed data extraction, high-efficiency bitwise algorithms, and robust testing with PyTest.

TECHNICAL SKILLS

- **Python Core & Frameworks:** Python 3.12+, FastAPI, Uvicorn, Asyncio, Subprocess Isolation, Tree-Sitter AST, ChromaDB, PyTest
- **Data Pipelines & Automation:** Playwright, Puppeteer, Selenium, Automated Scraper Pipelines, CSV/JSON Processing, SQLite Ingestion, Data Transformation
- **Backend & Distributed Systems:** RESTful API Design, Microservices, Redis 7 (LRU), Turso LibSQL, PostgreSQL, MongoDB, Docker
- **Cloud Infrastructure:** Oracle Cloud (OCI ARM64), Docker Compose, Caddy Reverse Proxy, GitHub Actions CI/CD, Linux Systems
- **Languages & Web:** Python, TypeScript, JavaScript, C#, Java, SQL, React, HTML5/CSS3
- **Security & Testing:** PyTest, Vitest, xUnit, Web Crypto API, JWT Authentication, Zod/Pydantic Validation, Code Quality

TECHNICAL PROJECTS

Astra Vision - Developer Sandbox & Code Graph Parser (FastAPI, Python, Monaco, Tree-Sitter) Live Demo | GitHub

- **Product Architecture:** Architected an automated code-parsing platform and browser sandbox that allows developers to safely execute untrusted code while visualizing full-stack repository dependency graphs.
- **Sandbox Runtime Challenge:** Built a self-healing Python execution engine with subprocess isolation and security checkpoints to intercept unauthorized OS file system calls and network socket requests in real time.
- **Indexing Performance:** Integrated Tree-Sitter AST compilers and ChromaDB vector search to parse repository syntax nodes, enabling instant code graph dependency analysis under 200ms.

AstreWork — Autonomous Career Intelligence Engine & Distributed Harvester (OCI, Docker, Redis) Live Demo | GitHub

- **Creator & Lead Backend Developer:** Engineered distributed scrapers on Oracle Cloud (OCI ARM64) aggregating 14,000+ remote opportunities daily across 7 platforms with 0% duplication.
- **Cloud Infrastructure & Microservices:** Deployed on Oracle Cloud (OCI Ampere ARM64) within multi-stage Docker containers managed by Watchtower CI/CD and Caddy HTTP/3 reverse proxy with automated SSL, maintaining 99.9% uptime and sub-50ms global API latency.
- **Distributed Caching & Concurrency:** Architected data persistence across Turso Cloud LibSQL (distributed SQLite WAL) and Redis 7 LRU caching, cutting duplicate job ingestion to 0% and reducing repeated query latency by 85%.

Alyra Lock - Secure Zero-Knowledge Password Vault (React, TypeScript, Express, MongoDB) Live Demo | GitHub

- **Product Architecture:** Engineered a client-side zero-knowledge password vault ensuring master encryption keys remain entirely isolated inside user browser memory, mitigating cloud database leak risks.
- **Cryptographic Architecture:** Implemented Web Crypto API primitives utilizing AES-GCM (256-bit) payload encryption and PBKDF2 key derivation over 100,000 hashing iterations for key security.
- **Performance & Security:** Developed high-throughput Express.js REST API sync endpoints with JWT authentication and strict CORS headers, keeping vault database sync latency under 200ms.

TECHNICAL SIMULATIONS & VIRTUAL EXPERIENCES

Commonwealth Bank (CommBank) - Software Development Virtual Simulation June 2026

- **Problem & Scope:** Diagnosed and resolved silent data overwrite bugs across high-traffic C#/.NET Core financial API controllers handling MongoDB document updates across financial services endpoints.
- **Action Taken:** Extended C# Web API controllers with $set atomic operators for partial payload updates and modernized an interactive React/Redux Goal Manager UI component for banking clients.
- **Key Outcome:** Authored comprehensive automated test suites using xUnit and Moq, covering complex boundary conditions to guarantee high reliability and transactional consistency.

Y Combinator (YC Startup - Shiptivity) - Software Development Virtual Simulation June 2026

- **Problem & Scope:** Fixed key constraint collisions and slow re-rendering performance on a real-time drag-and-drop Kanban workflow board used by engineering teams to track task state transitions.
- **Action Taken:** Architected dynamic SQLite priority reordering logic using atomic transactions to update priority ranks sequentially and integrated Dragula drag-and-drop state hooks in React.
- **Key Outcome:** Reduced UI component re-render cycles by 30% and patched OpenSSL compilation bottlenecks in legacy Node.js Webpack configurations to optimize runtime builds.

Walmart USA - Advanced Software Development Virtual Simulation May 2026

- **Problem & Scope:** Solved indexing overhead and performance bottlenecks in high-volume retail inventory processing queues handling unnormalized data ingestion from legacy spreadsheets.
- **Action Taken:** Implemented a generic K-ary Max Heap structure in Java using fast bitwise shift operators (<<, >>>) and built 3NF database schemas with design pattern abstractions.
- **Key Outcome:** Accelerated queue indexing calculations by 35% and engineered automated Python ETL pipelines using csv and sqlite3 to clean and consolidate disparate spreadsheets.

EDUCATION

Bachelor of Computer Applications (BCA) - Amity University Noida Expected July 2028

Relevant Coursework: Data Structures & Algorithms (DSA) • Database Management Systems (DBMS) • Operating Systems (OS) • Computer Networks • Object-Oriented Programming (OOP)